



BUILT FOR HANDS THAT ARE BUSY

Gloves on. Hands full. Just speak.

Speak a goods receipt. Review and confirm it. It posts into **SAP** — a mock S/4HANA behind SAP's released OData API.



RECEIVING DOCK • MORNING SHIFT

Lena has both hands under a box.

The pallet landed four minutes ago and it is on no one's books yet. Until that delivery is a document in SAP, the company does not know it owns the goods — purchasing chases it, production plans without it.

Booking it costs Lena a walk: gloves off, across the floor, to a shared terminal. Every delivery, every shift.

WHAT THIS TOUCHES — AND WHAT IT DOES NOT

A shift is a bundle of tasks.

Unload the pallet	UNTOUCHED	Type the material number	GONE
Check the packing slip	UNTOUCHED	Look up which bin it goes to	AUTOMATIC
Set damaged cartons aside	UNTOUCHED	Post the goods receipt	BY VOICE
Take the gloves off	GONE	Correct a wrong posting	BY VOICE
Walk to the shared terminal	GONE	Hear the line before it is true	NEW

GlovesOn cannot unload the pallet. It removes the three tasks that exist only because the record lives on a screen — and adds one worth doing.

SAY IT. IT POSTS.

Both hands on the box. The receipt **posts into SAP.**

YOU "Twenty pieces of material 4711 arrived."
GLOVESON "Twenty pieces of Hex Bolt M8x40 Zinc Plated into bin A-03-02 – confirm?"
YOU "Actually twelve."
GLOVESON "Twelve pieces ... into bin A-03-02 – confirm?"
YOU "Yes."

→ **Material document 4937706921 · movement 501 · stock 271 → 283**

A mock S/4HANA that speaks SAP's released OData contract. One setting from a live tenant. Hands never leave the box.

REAL SCREEN · SESSION OF 15 SEPTEMBER 2026

A result you can review.

GOODS RECEIPT · MATERIAL DOCUMENT · MOCK S/4HANA

Document **4937706921** ✓ assigned 13:51:56

POSTED · 501

RECORDED · 13:51:56 stock 271 → 283 EA

12 EA of Hex Bolt M8x40 Zinc Plated are in bin A-03-02.

Wrong? Say "reverse it". A reversal takes them back out and both documents stay.

Material ✓
4711

Description ✓ material master
Hex Bolt M8x40 Zinc Plated

Quantity ✓ confirmed
12 EA

Storage bin ✓ the material's own
A-03-02

Plant / location ✓ default
1000 / 0001

Stock ✓ from the same result
271 → 283 EA

WHY THIS DOCUMENT EXISTS
"Twelve pieces of Hex Bolt M8x40 Zinc Plated into aisle A, level three, position two." — as the agent reported it, 13:51:56.

✓ Receipt saved

Material document **4937706921**, movement **501**, stock **271 → 283** — recorded through SAP's released OData path into the demo's mock S/4HANA, after the worker corrected twenty to twelve during the read-back.

The read-back names the **description**, not just the number — which is what lets a worker catch they are holding nuts, not bolts.

The card carries **the sentence that caused the document**. An earlier run, document 4922857164 with its refused duplicate and its reversal, is kept as JSON in receipts/.

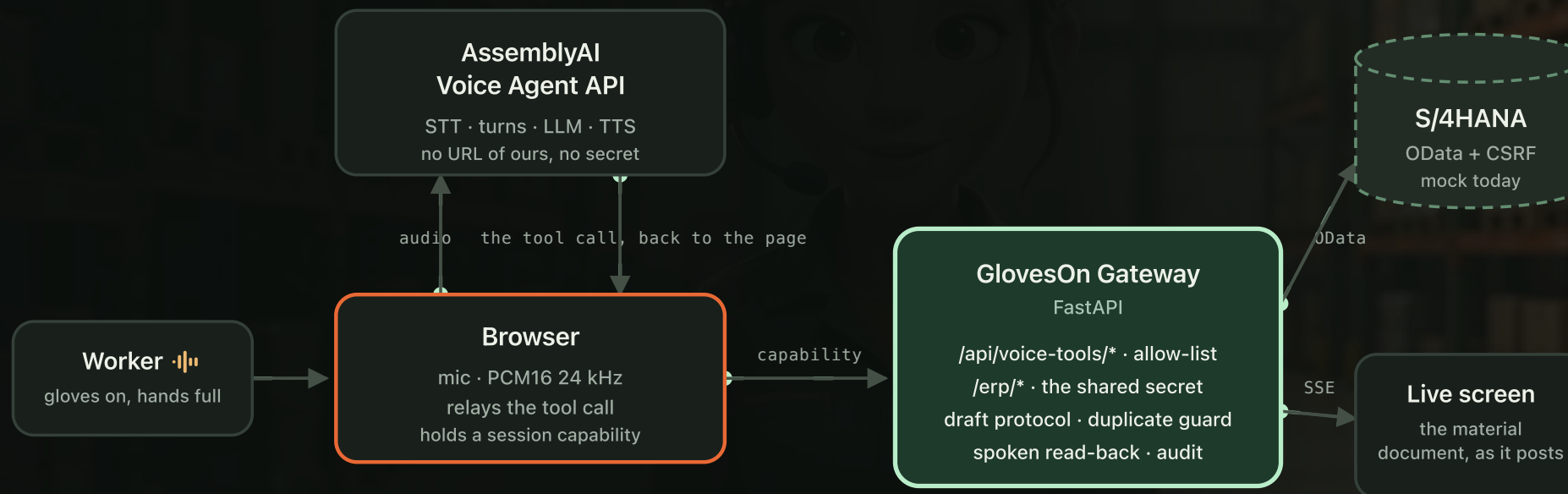
ASKING IS EASY. FINISHING IS THE HARD PART.

Five guardrails, each one deliberate.

- 1 The read-back names four things.**
Quantity, unit, material description, bin. The description is load-bearing.
- 2 Writes require a confirmed draft.**
The gateway validates the exact draft. Spoken consent is reported by the agent; results appear only after the tool returns.
- 3 A duplicate is refused.**
The same receipt within two minutes is caught — by a guard in the gateway, because real S/4HANA would accept it twice.
- 4 Nothing is deleted — it is reversed.**
A wrong receipt gets a reversal document, 102 against a 101. Both stay, because that is what a ledger is.
- 5 Every document answers why it exists.**
Click a posted row: the sentence the worker confirmed, and the second it happened.

THE AGENT NEVER SEES SAP

One boundary carries every guardrail.



the agent never sees SAP, and never sees the gateway either — the write is the gateway's alone

It speaks in warehouse terms; the gateway owns auth, SAP field translation and every write guardrail. The mock sits opposite the gateway, reached like any other system — so **SAP_BASE_URL** pointed at a real tenant is the whole migration. No second code path.

THE WHOLE VOICE LOOP IS THE VOICE AGENT API

One connection: speech, reasoning, voice, tools.

Streaming **speech-to-text**

LLM routing

Voice output

Turn-taking & VAD

JSON-Schema client functions

Browser-to-gateway tool bridge

transcription_prompt

39 keyterms → SAP material numbers

voice_focus: far-field

Barge-in

session.resume

Session token minted server-side

Turn detection is pinned (**800 / 2000 ms**) for non-native speakers — adaptive pacing is deliberately off, so we do not claim per-speaker adaptation.

EVERYONE ELSE VOICE-GUIDES PLANNED WORK

The unplanned pallet has no voice path.

THE INCUMBENTS

Honeywell · Lydia · Ivanti

The WMS issues the task; the worker answers fixed prompts. Decades of it. But a worker cannot initiate a document by describing it.

SAP ITSELF

Joule

Posts and reverses goods movements — by **chat**, today. Real-time voice announced for its top cloud tiers, GA planned H2 2026.

GlovesOn is the unplanned moment — by voice, today.

AN ESTABLISHED BUDGET LINE

Zero per-seat hardware against a \$5,000 norm.

\$4.8–6.5B

voice-directed warehousing today · ~14–17%/yr (analyst estimates, cited as a range)

~\$5,000

incumbent cost per user, hardware + software

\$0.00

GlovesOn per-seat hardware — a browser and a headset. Voice: \$0.075 per connected minute.

No ROI figure is quoted — session length per receipt is unmeasured (docs/nfr.md §5).

NAMED, NOT HIDDEN

What this does not do yet.

- **The ERP is a mock** — the OData contract and the CSRF handshake are real; a live tenant was tried and the Business Accelerator Hub sandbox answered **401** on all four required services (10 September). Nothing here has run against real S/4HANA.
- **No employee sign-in or SAP principal propagation** — session capabilities protect tool calls, but do not establish a worker's identity.
- **Noise robustness is unmeasured** — voice_focus: far-field is set, but a config change is a claim, not a measurement.
- **Session events are isolated; warehouse records are shared** — this is a demonstration, not a tenant-isolated production system.
- **The supported product language is English**; other languages are outside the demo scope.

Naming the gap is the credibility. Every one is written up in the repo, and the judge guide ends with them.

WHAT A JUDGE CAN CHECK

Posts the goods receipt **straight into SAP** — here a mock S/4HANA that speaks SAP's released OData contract, one setting away from a live tenant.

WHAT IT DOES

- `X-CSRF-Token`: Fetch handshake, `A_MaterialDocumentHeader`, movement types 101 / 102
- Read-back names quantity, unit, material *description* and bin; nothing posts without a spoken yes
- Wrong receipt is reversed, never deleted — both documents stay
- The same receipt twice is refused in the gateway

WHAT YOU CAN CHECK

- **gloveson.space** — no login, talk to it
- `github.com/kosesena/GlovesOn` — **281 commits** since 5 Sept, **97 tests**, CI green, MIT
- `receipts/` — **13 steps** of one real run: document **4922857164** posted, the same receipt refused, then reversed
- `/api/provenance/4922857164` — the sentence that caused the document
- `docs/adr/` — six decisions, each with the option rejected and its cost

NOT VERIFIED

- Real S/4HANA: the sandbox answered **401** on all four required services
- The posting is not made as the worker — no principal propagation yet
- Accents and forklift noise: not benchmarked
- Email, calls and notes are demo records; nothing reaches a real person

GLOVESON

It doesn't just stop and ask. It finishes — and leaves a document that can tell you **why it exists.**

Live demo · gloveson.space

github.com/kosesena/GlovesOn

Judge guide — every claim → a file & a command

Gloves on. Hands full. **Just speak.**

Built on the AssemblyAI Voice Agent API

Demo: mock ERP · simulated calls · demo email outbox